Octopus: Alleviating Hallucination via Dynamic Contrastive Decoding

Anonymous CVPR submission

Abstract

Large vision-language models (LVLMs) still hallucinate objects, attributes, and relations that are weakly supported or absent in the image. Recent mitigation methods based on contrastive decoding (CD) improve faithfulness at test time, but each strategy targets only a subset of hallucination cases and applying a single strategy globally is therefore sub-optimal. We present Octopus, a plug-and-play controller that selects the most suitable CD strategy at each generation step. Octopuskeeps the LVLM frozen and adds a lightweight transformer encoder with a learnable decision token that reads the current hidden states and predicts one of four actions: apply VCD, M3ID, AVISC, or take no action. The selected strategy then perturbs the decoding distribution for the current token only. The controller is trained with a reference-free DPO-style objective on preferred versus dispreferred action sequences, enabling policy learning without retraining the backbone LVLM. On LLaVA-1.5-7B, Octopusimproves MMHalBench score from 1.59 to 2.61 and reduces its hallucination rate from 0.72 to 0.50, lowers AMBER CHAIR from 8.0 to 4.8, and raises AMBER discrimination accuracy/F1 from 67.0/71.1 to 76.7/82.7. The gains transfer to InstructBLIP, where Octopusimproves MMHalBench from 1.84 to 2.31 and reduces AMBER CHAIR from 8.4 to 6.1. These results show that dynamic, token-level CD selection is an effective and deployable way to reduce LVLM hallucination.

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1. Introduction

Large vision-language models (LVLMs) such as LLaVA [14, 15] and InstructBLIP [3] have become strong multimodal assistants, but they still generate text that is only weakly grounded in the image. Hallucinations remain visible on captioning-style benchmarks, discriminative benchmarks, and human-centered evaluations such as MMHalBench [25], POPE [12], and AMBER [28]. This gap has motivated both retraining-based alignment methods [21, 32, 33, 35, 36] and training-free decoding methods [4, 5, 7, 10, 18, 29, 31].

Among test-time solutions, contrastive decoding is particularly attractive because it is plug-and-play: a base LVLM is queried with an original input and a hallucination-amplifying variant, and the difference between the two distributions suppresses hallucinated tokens [10, 29, 31]. However, existing CD methods are largely one-size-fits-all. VCD [10] mainly counters language priors by contrasting original and distorted images. M3ID [4] reinforces visual grounding when visual information is being forgotten during generation. AVISC [31] specifically targets attention bias toward visually uninformative tokens. These methods are complementary, not redundant.

Our central observation is that hallucination is hybrid at two levels. First, across samples, different CD methods are effective on different subsets of examples. Fig. 3 shows that on both generative and discriminative tasks, the largest portion of effective cases is covered by only one strategy, while all three strategies are simultaneously effective on only 7% to 16% of cases. Second, within a single generated response, different tokens can fail for different reasons. Fig. 5 illustrates a single hallucinated description in which the erroneous tokens "sitting", "lying", and "person" align with different failure modes. A fixed strategy therefore wastes capacity: even when a sample benefits from CD, it does not necessarily benefit from the same CD strategy for every token.

We address this mismatch with Octopus, a dynamic controller for token-level CD selection. Octopusleaves the backbone LVLM frozen and adds a small policy network on top of its hidden states. At each decoding step, a learnable decision token, implemented as a `cls` token in the released code, attends to the current hidden states and predicts one of four actions: VCD, M3ID, AVISC, or no action. The chosen action activates one "tentacle" for that token and only that token. The result is a dynamic workflow that adapts hallucination mitigation to the current multimodal context without modifying the backbone model.

We train the controller with a DPO-style preference objective [19]. The released pipeline samples multiple action sequences offline, identifies preferred and dispreferred sequences for each example, and stores them as winner/loser trajectories consumed by the training loader. The controller then learns to assign higher probability to the preferred action workflow while the LVLM stays frozen. This preserves deployability and keeps the optimization problem small.

Our contributions are threefold. First, we provide em-

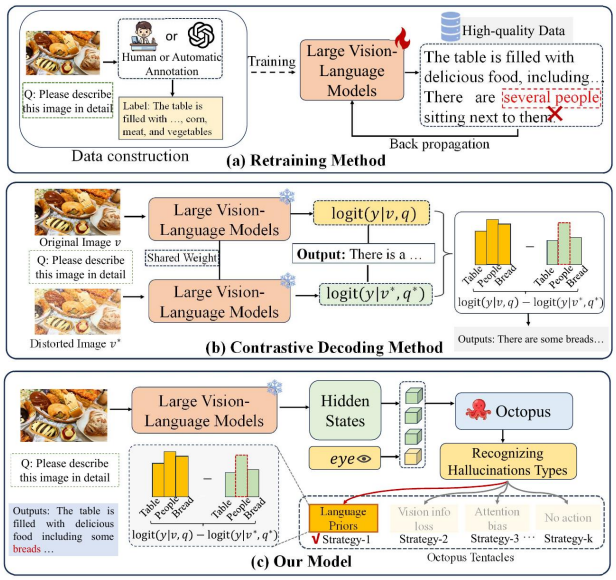

Figure 1. Paradigm comparison. Retraining methods modify LVLM parameters using additional supervision; standard contrastive decoding applies one manually chosen strategy to the entire sample; Octopuspredicts the suitable hallucination-mitigation action from the current hidden states and can switch strategies across tokens.

pirical evidence that hallucination causes are hybrid across both samples and generation steps. Second, we introduce Octopus, a lightweight token-level controller for dynamic CD selection over existing decoding strategies. Third, we show strong improvements on both generative and discriminative hallucination benchmarks, with good transfer from LLaVA-1.5-7B to InstructBLIP.

2. Related Work

2.1. Hallucination in LVLMs

Hallucination in multimodal generation has been studied from both the modeling and evaluation perspectives. Early analyses emphasized object hallucination in captioning [20] and later showed that instruction-tuned LVLMs also hallucinate under open-ended prompts and polling-style queries [12]. Benchmarking work broadened the picture from object-only errors to richer hallucination types. MMHalBench [25] stresses human-aligned factuality under challenging prompts, while AMBER [28] provides an LLM-free evaluation covering generative and discriminative hallucination, including object, attribute, and relation errors.

2.2. Retraining-Based Mitigation

One line of work reduces hallucination through additional training or preference tuning. Examples include modality alignment with preference fine-tuning [36], hallucination-aware DPO [35], contrastive tuning with augmented hallucinated pairs [21], fine-grained correctional feedback [33], and sentence-level AI feedback [32]. These methods can be effective, but they require new supervision, backbone updates, or both. In contrast, Octopuskeeps the LVLM frozen and learns only a small controller.

2.3. Training-Free Decoding

Recent training-free methods intervene at inference time. VCD [10] contrasts original and distorted images to offset language priors. ICD [29] perturbs the instruction side. AVISC [31] identifies blind image tokens and calibrates their influence through contrastive decoding. OPERA [5] penalizes over-trusted summary tokens during beam search, while LCD [16], SID [7], and ConVis [18] provide additional decoding-based alternatives. Our method is complementary to this literature: we do not invent a fourth handcrafted CD rule, but instead learn when to invoke existing ones.

2.4. Preference Optimization for Multimodal Models

Preference optimization has become a standard alternative to RLHF [17, 19, 24]. It has also been adopted in multimodal settings [11, 27]. Octopususes preference learning differently: the optimized policy is not the LVLM itself, but a small action selector operating over decoding strategies. This makes preference learning lightweight and directly targeted at token-level mitigation behavior.

3. Method

3.1. Problem Setup

Let an LVLM receive an image-question pair (v, q) and generate a response autoregressively. At step t, the frozen LVLM produces hidden states $H_t = [h_0, \ldots, h_t]$ and a base next-token logit vector z_t. Existing CD methods define one perturbed distribution per strategy by contrasting the original context with an auxiliary context, such as a distorted image or a masked-attention variant. Standard practice commits to one strategy for the whole sample. We instead learn a policy that selects the most suitable strategy at each step.

3.2. Why Dynamic Selection?

We motivate token-level control from two empirical observations.

First, at the sample level, different CD methods cover different subsets of examples. Fig. 3 shows that on AMBER, Object HalBench, and MMHalBench, only one strategy is effective for 63%, 65%, and 48% of effective samples, respectively. The same pattern appears on discriminative tasks from GQA [6], A-OKVQA [23], and COCO [13],

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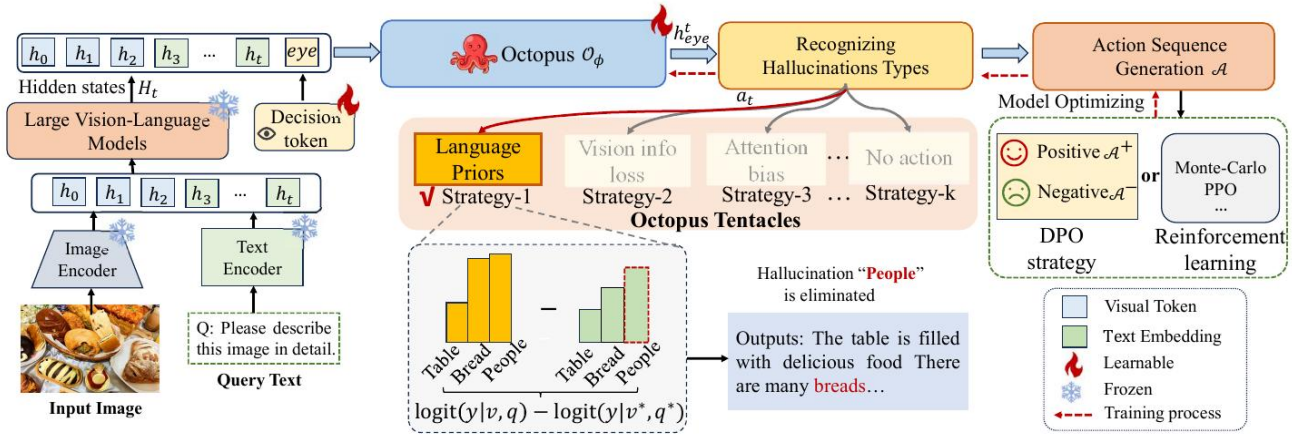

Figure 2. Overview of Octopus. The frozen LVLM produces hidden states at each generation step. A lightweight controller reads these states together with a learnable decision token (the "eye") and predicts an action over the candidate tentacles: VCD for language-prior errors, M3ID for visual-information loss, AVISC for attention bias, or no action. The selected tentacle modifies the current-token logits through its original CD rule. The released code trains only the controller parameters with a reference-free DPO-style objective on preferred and dispreferred action sequences.

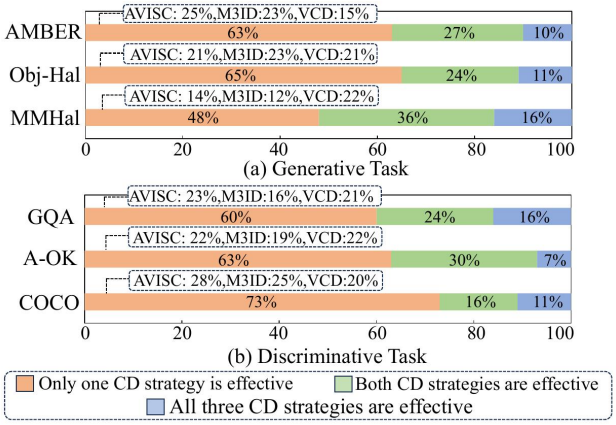

Figure 3. Sample-level motivation. Across both generative and discriminative tasks, the largest fraction of effective cases is covered by only one CD strategy, while full agreement among all three strategies is rare. This motivates adaptive selection rather than a global fixed strategy.

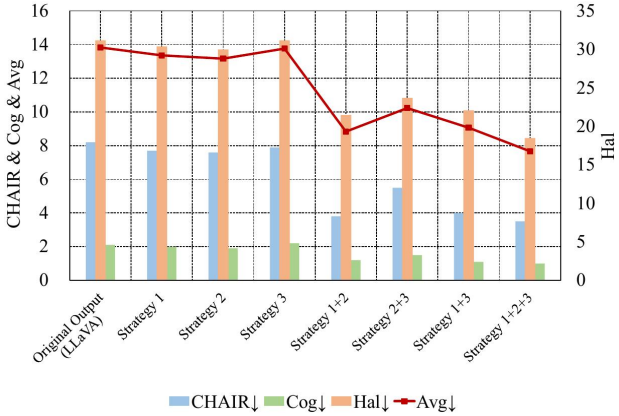

Figure 4. Token-level motivation. The best performance improves as more strategies become available across generation steps, supporting the claim that different tokens may require different mitigation actions.

where the "only one strategy" portion ranges from 60% to 73%. Since the overlap among strategies is limited, a global fixed choice is inherently brittle.

Second, the failure mode can change inside one response. Fig. 4 shows that allowing more strategies across tokens monotonically improves the best achievable hallucination metrics, and Fig. 5 visualizes how different hallucinated tokens align with language priors, visual-information loss, and attention bias. These observations motivate a token-level action policy rather than a sample-level routing rule.

3.3. OctopusController

Fig. 2 summarizes the architecture. We add a lightweight controller π_ϕ on top of the frozen LVLM hidden states. In the released implementation, the controller is a two-layer transformer encoder [26] followed by an MLP classifier. A learnable decision token e is concatenated to the hidden-state sequence,

$$\hat{H}_t = [e; H_t], \tag{1}$$

and passed through the controller. The classifier reads the transformed decision token and produces a categorical dis-

tribution over four actions:

$$\pi_\phi(a_t \mid H_t), \quad a_t \in \{\texttt{none}, \texttt{VCD}, \texttt{M3ID}, \texttt{AVISC}\}. \quad (2)$$

At inference, Octopuschooses the highest-probability action and applies the corresponding tentacle only for the current step.

This design has two desirable properties. First, the controller sees the current multimodal state through the LVLM hidden states, so it can react to the evolving decoding context. Second, the controller is small: in the released code, only the decision token, transformer encoder, and MLP are optimized, while the LVLM backbone and language head are frozen.

3.4. Octopus Tentacles

Octopusreuses existing CD strategies as actions.

VCD. Visual Contrastive Decoding [10] contrasts the original image with a distorted image to suppress language-prior-driven hallucinations.

M3ID. Multi-Modal Mutual-Information Decoding [4] amplifies visually grounded tokens and mitigates the tendency to drift toward pure language priors later in generation.

AVISC. Attentional Vision Calibration [31] identifies blind tokens from attention patterns and calibrates them through a contrastive forward pass, addressing attention bias toward uninformative visual regions.

Null action. When none of the contrastive strategies is useful, Octopusleaves the base LVLM distribution unchanged. This is important because over-intervention can hurt response quality on already well-grounded tokens.

For VCD and AVISC, the implementation follows the standard contrastive form

$$\tilde{z}_t = (1+\alpha)z_t - \alpha z_t^-, \quad (3)$$

with adaptive plausibility masking controlled by β before sampling. For M3ID, the implementation uses the original mutual-information-inspired update from [4]. Octopusitself does not modify these rules; it only decides when each one should be activated.

3.5. Training with Preference Signals

The controller is trained from preferred and dispreferred action sequences. The released data generator samples multiple stochastic action trajectories per example and records their responses. Offline evaluation then identifies a winner sequence a^+ and a loser sequence a^-, which are stored in the training JSON and loaded as `y_w` and `y_l`. The controller is trained with a reference-free DPO-style loss:

$$\mathcal{L} = -\log \sigma\left(\beta\left[\log \pi_\phi(a^+ \mid x) - \log \pi_\phi(a^- \mid x)\right]\right), \quad (4)$$

where x is the image-question input and β is a temperature parameter. This matches the released implementation, which omits a separate reference policy and optimizes the controller directly.

Algorithmically, training proceeds as follows: (1) run the frozen LVLM with the controller in sampling mode to obtain action probabilities at each step; (2) gather log-probabilities of the preferred and dispreferred action sequences; (3) optimize the DPO-style loss; and (4) update only controller parameters with Adam [9]. Because the backbone is frozen, training is substantially cheaper than retraining the whole LVLM.

4. Experiments

4.1. Experimental Setup

We evaluate on two LVLM backbones: LLaVA-1.5-7B [14] and InstructBLIP [3]. Our candidate tentacles are VCD [10], M3ID [4], and AVISC [31], plus a null action. The controller is the released two-layer transformer encoder with hidden size matched to the LVLM hidden dimension, two attention heads, and a four-way output classifier. Training uses the released preference-learning code with Adam at learning rate 10^{-5} for 5 epochs, DPO temperature $\beta = 1.0$, and contrastive plausibility parameter `cd_beta`=0.1. During inference, decoding is stochastic with temperature 1.0, top-p 1.0, and up to 64 generated tokens.

We report results on MMHalBench [25], Object HalBench/POPE-style object hallucination evaluation [12], AMBER [28], and POPE-MSCOCO [12]. Following the provided tables, we separate *referenced results* from *comparable results*: the former are copied from prior papers and are not directly comparable to our reproduced runs, while the latter are measured under a shared evaluation pipeline.

4.2. Generative Results

Table 1 compares Octopuswith prior methods on generative tasks. On LLaVA-1.5-7B, Octopusimproves MMHalBench score from 1.59 to 2.61 and reduces the hallucination rate from 0.72 to 0.50. On Object HalBench, it lowers CHAIR_s from 25.0 to 20.8 and CHAIR_i from 9.2 to 6.6. On AMBER captioning, it reduces CHAIR from 8.0 to 4.8, increases coverage from 44.5 to 49.2, lowers hallucination rate from 31.0 to 23.4, and reduces cognition error from 2.2 to 1.2.

The gains are consistent across reproduced CD baselines. Relative to the strongest reproduced single-strategy method on LLaVA-1.5-7B, Octopusimproves MMHalBench score by 0.42 over AVISC, lowers Object HalBench CHAIR_i by 0.7 over M3ID, and lowers AMBER CHAIR by 1.2 over M3ID. This pattern supports our claim that no single CD strategy is uniformly best.

The policy also transfers to InstructBLIP. MMHalBench score rises from 1.84 to 2.31 and hallucination rate drops from 0.64 to 0.49. On AMBER captioning, CHAIR falls

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from 8.4 to 6.1 and hallucination rate from 31.1 to 22.2. These improvements indicate that the controller is not tied to one specific backbone.

4.3. Discriminative Results

Table 2 reports discriminative performance on AMBER and POPE-MSCOCO. On LLaVA-1.5-7B, Octopusimproves AMBER discrimination accuracy/F1 from 67.0/71.1 to 76.7/82.7. On POPE-MSCOCO, it reaches 85.79/83.44 on the aggregated split, improving over the base model by 3.75 accuracy points and 3.02 F1 points. On InstructBLIP, Octopusimproves AMBER discrimination from 68.2/74.6 to 74.0/79.7 and reaches 84.79/83.43 on POPE-MSCOCO ALL.

Importantly, the same controller helps on both open-ended generation and binary-style discrimination. This suggests that Octopusis not merely improving fluency or verbosity; it is consistently steering the model toward better visual grounding.

5. Analysis

5.1. Dynamic Selection Matters

Table 3 ablates the action space on AMBER captioning. Using multiple fixed CD strategies without the controller already improves over the plain baseline (for example, CHAIR 8.0 to 6.9), confirming that strategy diversity helps. However, the full controller with all three tentacles performs best, reaching CHAIR 4.8, coverage 49.2, hallucination rate 23.4, and cognition error 1.2. This shows that simply exposing the model to multiple strategies is insufficient; a learned token-level selector is needed to exploit them effectively.

5.2. Controller Design Choices

Table 4 studies several design alternatives. The final configuration achieves the best CHAIR and cognition score, while remaining competitive on coverage and hallucination rate. Compared with simple scoring heuristics (Cover and Average), alternative optimization strategies (Monte Carlo and PPO [22, 30]), and simpler feature aggregators such as pooling [2] or cross-attention [26], the released configuration provides the best overall trade-off. This supports the use of the decision-token transformer encoder trained with preference supervision.

5.3. Qualitative Evidence

Fig. 6 compares responses from the base LLaVA-1.5 model, individual CD strategies, and Octopus. The base model hallucinates additional people in a soccer image. Each single strategy reduces the error only partially and still invents extra people or uncertain roles. Octopusremoves the hallucinated entities and produces a grounded description focused on the visible player, referee, and field marker.

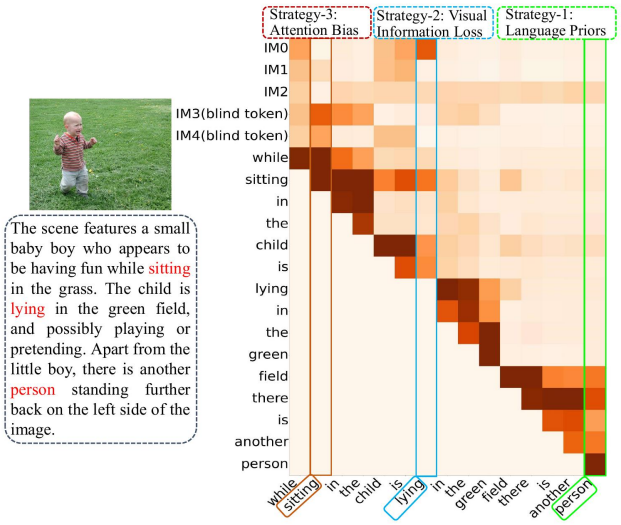

Figure 5. Token-level qualitative analysis. Different hallucinated tokens within one response align with different causes, supporting the need for per-token strategy switching.

Fig. 5 offers token-level intuition. In the illustrated example, the hallucinated tokens "sitting", "lying", and "person" are aligned with different failure modes: attention bias, visual-information loss, and language priors. This supports the main premise of the paper that token-level adaptation is necessary.

5.4. Discussion

Octopusis attractive because it separates *what* to mitigate from *how* to mitigate. The tentacles can be any existing or future decoding strategy, while the controller learns the routing policy. This makes the framework extensible: adding a new test-time hallucination strategy only enlarges the action space and does not require retraining the backbone LVLM.

At the same time, our results should be interpreted with the evaluation protocol in mind. Some prior methods in Table 1 are listed as referenced rather than directly comparable. Our strongest claims therefore concern the controlled comparisons against the reproduced baselines and the consistent gains across both LLaVA-1.5-7B and InstructBLIP.

6. Conclusion

We presented Octopus, a dynamic contrastive decoding framework for hallucination mitigation in LVLMs. The key idea is to replace a fixed global CD strategy with a token-level controller that selects among VCD, M3ID, AVISC, and a null action from the current hidden states. The controller is lightweight, keeps the LVLM frozen, and is trained with a reference-free DPO-style objective over action-sequence preferences. Experiments across MMHal-

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Table 1. Comparison with the state-of-the-art methods for the generative task across three datasets. For reference, we also provide the results of fine-tuning LVLMs. † signifies results reproduced with the official implementation codes.

LVLMs	Method	MMHalBench		Object HalBench		AMBER			
		Score ↑	HalRate ↓	$CHAIR_s$ ↓	$CHAIR_i$ ↓	CHAIR ↓	Cover. ↑	HalRate ↓	Cog. ↓
Referenced Results (Not Directly Comparable)									
GPT-4V [1]	Base	3.49	0.28	13.6	7.3	4.6	67.1	30.7	2.6
LLaVA-v1.5 -7B[14]	HACL [8]	2.13	0.50	-	-	-	-	-	-
	POVID [36]	2.08	0.56	48.1	24.4	-	-	-	-
	EOS [34]	2.03	0.59	40.3	17.8	5.1	49.1	22.7	2.0
	HA-DPO [35]	1.97	0.60	39.9	19.9	6.7	49.8	30.9	3.3
	HALVA [21]	2.25	0.54	-	-	6.6	53.0	32.2	3.4
Comparable Results									
LLaVA-v1.5 -7B [14]	Base	1.59	0.72	25.0	9.2	8.0	44.5	31.0	2.2
	+ LCD [16]	-	-	61.0	16.1	-	-	-	-
	+ ICD [29]	-	-	47.4	13.9	-	-	-	-
	+ OPERA [5]	2.15	0.54	45.1	22.3	-	-	-	-
	+ VCD [10]	1.96†	0.64†	23.6†	8.4†	6.7	46.5	27.8	2.0
	+ M3ID [4]	2.14†	0.61†	23.2†	7.3†	6.0	48.9	26.0	1.5
	+ AVISC [31]	2.19†	0.59†	22.1†	7.8†	6.3	46.6	25.6	2.0
	+ Octopus(Ours)	**2.61**	**0.50**	**20.8**	**6.6**	**4.8**	**49.2**	**23.4**	**1.2**
Instruct-BLIP [3]	Base	1.84	0.64	0.7	9.1	8.4	46.4	31.1	2.6
	+ LCD [16]	-	-	17.4	10.7	-	-	-	-
	+ ICD [29]	-	-	15.2	8.0	-	-	-	-
	+ OPERA [5]	-	-	16.6	6.8	-	-	-	-
	+ VCD [10]	1.75†	0.64†	0.8†	8.9†	7.6	47.7	29.9	2.2
	+ M3ID [4]	1.70†	0.65†	0.9†	7.6†	6.9	47.2	27.5	2.2
	+ AVISC [31]	2.03†	0.59†	0.7†	8.3†	6.7	46.7	28.0	2.6
	Octopus (Ours)	**2.31**	**0.49**	**0.5**	**6.8**	**6.1**	**48.5**	**22.2**	**1.3**

Table 2. Comparison with the state-of-the-art methods for the discriminative tasks across two datasets. † signifies results reproduced with the official implementation codes.

	AMBER		POPE_MSCOCO							
	Discrimination		Random		Popular		Adversarial		ALL	
	Accuracy	F1	Accuracy	F1	Accuracy	F1	Accuracy	F1	Accuracy	F1
LLaVA-1.5-7B[14]	67.00	71.10	83.77	81.94	82.57	80.86	79.77	78.47	82.04	80.42
+ICD [29]	-	-	**87.51**	83.28	83.15	83.91	79.13	80.41	83.26	82.53
+ConVis [18]	-	-	84.70	-	83.20	-	81.10	-	83.00	-
+OPERA [5]	-	-	84.40	-	83.40	-	81.20	-	83.00	-
+VCD [10]	67.30	71.10	85.43	83.99	83.17	81.94	80.27	79.49	82.96	81.81
+M3ID [4] †	67.25	70.90	86.13	81.85	82.07	80.77	79.50	78.15	82.57	80.26
+AVISC [31]	70.70	75.45	84.67	82.21	83.67	81.27	81.83	79.55	83.39	81.01
+Octopus (Ours)	**76.70**	**82.70**	**87.51**	**85.40**	**85.20**	**84.19**	**82.22**	**81.44**	**85.79**	**83.44**
InstructBLIP [3]	68.20	74.60	81.53	81.19	78.47	78.75	77.43	78.00	79.14	79.31
+ICD [29]	-	-	84.36	83.82	77.88	78.70	75.17	77.23	79.14	79.92
+OPERA [5]	-	-	84.57	83.74	78.24	79.15	74.59	76.33	79.13	79.74
+VCD [10]	69.65	75.90	82.03	81.56	79.13	79.20	77.23	77.72	79.46	79.49
+M3ID [4] †	69.05	75.25	82.33	81.53	80.90	80.42	78.53	78.49	80.59	80.15
+AVISC [31]	72.60	78.60	86.03	84.41	84.27	82.77	81.83	80.67	84.04	82.62
+Octopus (Ours)	**74.00**	**79.70**	**86.63**	**85.30**	**84.90**	**83.55**	**82.83**	**81.43**	**84.79**	**83.43**

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Q: Please Describe this image in detail.

LLaVA-1.5: The image features a woman wearing a green uniform playing soccer, kicking the ball with great skill. The soccer ball is visible just in front of her, with her foot hovering over it. There are three other people visible in the scene. One person can be seen on the left side.

VCD: The image captures a soccer game in action, with a woman playing the position of the soccer ball kicker. She is skillfully kicking the ball towards the goal, while the soccer referee closely watches her play. There are a few other people in the scene, possibly team.

M3ID: The image features a soccer field with several people engaged in a game. A young female player, number 14, is kicking a soccer ball, while an official stands behind her, watching her play. Another person, likely another referee or an observer, is also present in the scene.

AVISC: The scene features a soccer field with several people present. At the center, a woman dressed in green is preparing to kick a soccer ball, possibly warming up or practicing her skills. Another woman can be seen standing nearby, likely observing or providing guidance.

Octopus: The image captures a soccer game in progress, with a woman in green kicking a soccer ball in the grass. A yellow flag is present nearby, indicating the sidelines or boundaries of the field. An official or referee, standing next to the woman.

Figure 6. Qualitative comparison on a soccer example. The base model and individual CD methods still hallucinate extra people, while Octopusproduces a more faithful description. Hallucinated phrases are highlighted in red in the figure.

Table 3. **Ablation study.** We select different numbers of contrastive decoding methods as candidates to demonstrate the effectiveness of our approach. "Str1, Str2, Str3" indicate CD strategy VCD [10], M3ID [4] and AVISC [31] in Sec. 3.4, respectively.

	Str1	Str2	Str3	Octopus	CHAIR ↓	Cover. ↑	Hal. ↓	Cog. ↓
1					8.0	44.5	31.0	2.2
2	✓	✓	✓		6.9	46.2	26.1	2.2
3	✓	✓		✓	5.5	48.7	25.8	1.5
4	✓		✓	✓	5.7	48.2	25.3	1.5
5		✓	✓	✓	5.5	48.4	26.2	1.6
6	✓	✓	✓	✓	**4.8**	**49.2**	**23.4**	**1.2**

Table 4. The effects of different settings for our method.

		CHAIR ↓	Cover. ↑	Hal. ↓	Cog. ↓
1	Cover	5.4	**50.1**	26.1	1.2
2	Average	5.0	49.8	23.0	1.4
3	Monte Carlo [30]	5.1	48.1	23.9	1.3
4	PPO [22]	5.8	47.5	**23.0**	1.5
5	Pooling [2]	6.2	44.8	25.0	1.6
6	Cross-atteition [26]	5.1	49.0	24.2	1.3
7	Ours	**4.8**	49.2	23.4	**1.2**

Bench, Object HalBench, AMBER, and POPE-MSCOCO show that dynamic token-level CD selection consistently reduces hallucination and improves grounding on both LLaVA-1.5-7B and InstructBLIP.

References

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